Assignment No. 1

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Customer Feedback Categorization

(i) Real-world NLP Tool

Tool selected: MonkeyLearn
Application: Customer Feedback Categorization
MonkeyLearn is an NLP-based text analysis platform that classifies customer feedback into predefined categories such as Pricing, Product Quality, Delivery, Customer Service, and Product Features.

(a) Input Data

Sample customer feedback given to the tool:

No.	Customer Feedback
1	"The product quality is excellent."
2	"The delivery was very late."
3	"The price is too expensive."
4	"Customer support was very helpful."
5	"The product stopped working after one week."
6	"Delivery was quick and the packaging was good."
7	"Bahut mehenga hai, price kam karo."
8	"Delivery bahut late hui."

Snapshot: Take a screenshot showing the above feedback entered into the selected tool.

(b) System Response

The system categorizes the feedback according to its detected topic/sentiment.

Input	Predicted Category
Product quality is excellent	Quality
Delivery was very late	Delivery
Price is too expensive	Pricing
Customer support was helpful	Customer Service
Product stopped working	Quality
Delivery was quick	Delivery
Bahut mehenga hai	Pricing
Delivery bahut late hui	Delivery

Snapshot: Take a screenshot of the tool's classification/output.

(c) Successes and Failures

Successful cases:

- English sentences were generally classified correctly.
- Clear words such as price, delivery, quality, and support helped the system identify categories.
- Short and direct feedback produced better results.

Failure cases:

- Hindi-English mixed sentences (Hinglish) may be classified incorrectly.
- Regional expressions may not have enough context.
- Sarcasm can be difficult to detect.
- The same sentence may belong to more than one category.
- Indian cultural references may be misunderstood if they are absent from the training data.

(ii) Comparison of Two NLP Tools

For comparison, use MonkeyLearn and Google Cloud Natural Language with the same input dataset.

Feedback	MonkeyLearn	Google NLP	Error/Observation
Product quality is excellent	Quality	Positive/Quality-related	Correct
Delivery was very late	Delivery	Negative	Different categorization
Price is too expensive	Pricing	Negative	Context/category difference
Customer support was helpful	Customer Service	Positive	Category vs sentiment
Product stopped working	Quality	Negative	Different output type
Bahut mehenga hai	Pricing	Negative/uncertain	Hindi input
Delivery bahut late hui	Delivery	Negative/uncertain	Hinglish
Packaging was damaged	Delivery/Quality	Negative	Multiple possible categories

(b) Types of Errors

- **Grammatical errors** – Incorrect or incomplete sentences can reduce accuracy.
- **Semantic errors** – Same words may have different meanings depending on context.
- **Translation errors** – Hindi/Hinglish expressions may not be interpreted correctly.
- **Context loss** – Short feedback may not provide enough information.
- **Cultural misunderstanding** – Indian-specific expressions may be classified incorrectly.
- **Multi-category problem** – One review may discuss both delivery and quality.

(c)Cultural/Regional Examples

Input: "Bhai, delivery toh ekdum late thi." | **Expected category:** Delivery
Possible problem: A model trained mainly on formal English may fail to understand the informal Hindi-English expression.
Input: "Bahut mehenga hai yaar." | **Expected category:** Pricing
Observation: The system may recognize negative sentiment but fail to assign the specific Pricing category.

(iii) Adapting the Tool for Marathi

I would choose Marathi as the Indian language for adaptation.

(a) Required Linguistic Resources

The system would require:

- Large Marathi customer-feedback corpus & category-labelled reviews
- Marathi POS tagger, tokenizer, and language model
- Marathi sentiment-analysis dataset and stop-word list
- Marathi word embeddings & Marathi-English/Hinglish parallel dataset

Example training data:

Marathi Feedback	Category
"उत्पादनाची गुणवत्ता चांगली आहे."	Quality
"डिलिव्हरी खूप उशिरा झाली."	Delivery
"किंमत खूप जास्त आहे."	Pricing

(b) Socio-Linguistic Challenges

1. **Code-switching:** Users mix Marathi and English: "Delivery खूप late झाली."
2. **Dialects:** Different regions of Maharashtra use different vocabulary and phrasing.
3. **Script variation:** Marathi written in Devanagari vs. Roman script (e.g. "Delivery khup late zali").
4. **Informal language & spelling:** Slang, abbreviations, emojis, and phonetic spelling variations.

Conclusion

> Customer Feedback Categorization uses NLP to automatically organize customer reviews into categories such as Pricing, Quality, Delivery, and Customer Service. Comparing tools demonstrates that while standard English inputs are handled effectively, Hindi/Hinglish, code-switching, and regional expressions present notable classification challenges. For Marathi adaptation, a dedicated annotated corpus, POS taggers, language models, and support for Romanized Marathi are required.